Timing a possible encounter

antocciphysics | TEACHER KEY | CPA

Learning target: Separate orbital period from an encounter time.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 12; 3 course days later.*
No. Tangent velocity arrows change direction, so inward acceleration and net force are needed.

02. Explain a worked example (worked)
Month 14 matches; a simultaneous encounter requires nearby positions at the same time.

03. Look and explain (model)
No; the given times differ.

04. Build the solution (practice)
Years 8 and 16; no Earth location not given.

05. Use the idea (reason)
A bound is not a probability distribution; need joint timing/position uncertainty and encounter geometry.

06. Choose a method (mixed) *Review: Unit 2, Day 3; 12 course days later.*
Mass 4 kg; weight 24 N toward the source. Weight depends on field.

07. Exit check (exit)
Orbital phase/time reference and both positions/geometry; uncertainty also valid.

08. STRETCH: Apply it another way (stretch)
Example A at months 1, 5, 9 and B at 3, 7, 11; no coincidence in stated interval.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 12; 3 course days later.*
Both become 1/3; $a_c = v^2/r$ and $F_{net} = mv^2/r$.

02. Explain a worked example (worked)
Month 14 matches; a simultaneous encounter requires nearby positions at the same time.

03. Represent and interpret (model)
No; the given times differ.

04. Solve and justify (practice)
6 and 42 months. Ideal periods/geometry fixed, does not include uncertainty, 3 D miss distance, etc.

05. Analyze the reasoning (reason)
A bound is not a probability distribution; need joint timing/position uncertainty and encounter geometry.

06. Choose a method (mixed) *Review: Unit 2, Day 3; 12 course days later.*
Mass 4 kg; new weight 24 N.

07. Exit check (exit)
Days 13-14 overlap; possible timing consistency, not proof of collision without spatial geometry/uncertainty.

08. STRETCH: Apply it another way (stretch)
First common time 19 months (n=3, m=2). Ignores miss distance, changing periods and uncertainty; timing alone does not prove impact.